DISTRIBUTION SUBSTATIONS WITH EDGE GENERATION

Design Basis: Full Three-Layer Instrument

DBA-ES-DI-FC5: Distribution Facility Class with Edge Generation

Issued Under DBA-ES: Electric Sector Design Basis Framework (v0.9)

Grandparent: DBA-EN-SF: Energy Sector Framework (v0.8)

Version 0.4 — DRAFT

June 2026

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For review by DisCo engineers, distribution planners, microgrid operators, defense installation energy managers, NERC reliability analysts, and state regulatory staff.

Section 1 — Purpose, Scope, and Version Status

1.1 Purpose and Version Scope

This document is the DBA-ES-DI-FC5 facility class instrument for Distribution Substations with Edge Generation. Version 0.3 completes the full DB→DBA→DBA-MA three-layer progression mandated by DBA-EN-SF §3.1. Version 0.1 established the DB layer. Version 0.2 added the DBA consequence envelope, the MA extension, and the recovery architecture. This version supersedes v0.2 in all respects.

This instrument applies the full DB→DBA→DBA-MA three-layer progression mandated by DBA-EN-SF §3.1. The non-adversarial Design Basis (DB layer) is established in Section 4. The Design Basis Accident consequence envelope (DBA layer) is established in Section 3.2. The Military Action Extension (DBA-MA layer) is developed in the scenarios of Section 5. All three layers are present in this instrument.

1.2 Instrument Type and Applicability

This is a facility-class instrument (FC instrument) issued under the DBA-ES Electric Sector Design Basis Framework. It applies to distribution substations that satisfy all three criteria established in Section 2.1: HV/EHV transmission input (69 kV and above), medium-voltage distribution output (4–25 kV), and on-site edge generation with an islanding architecture that supports intentional islanding under Command Broker authorization.

1.3 Relationship to Governing Documents

This document is issued under the DBA-ES Electric Sector Framework (v0.9), which is itself issued under the DBA-EN-SF Energy Sector Framework (v0.8) and the DBA-MA Parent Framework (IAEA L1 v1.4). The threat taxonomy, bounding methodology, consequence analysis structure, and adequacy standards are inherited from the sector framework. This facility-class document adds the FC5-specific physical configuration, asset taxonomy, bounding events, and consequence chains.

Series Grandparent: DBA-EN-SF: Energy Sector Framework (v0.8)

Series Parent: DBA-ES: Electric Sector Design Basis Framework (v0.9)

Predecessor document: DBA-ES FC5 and CntlCo Gap Analysis (v0.1, June 2026). Established facility class architecture, postulated event taxonomy, and preliminary design basis decisions carried forward into this instrument.

The Command Broker series (CB-Framework Cross-Sector v0.1, CB-UC-1 v0.1, Command Broker Implementation Guide v0.1) governs the Bright Line and Command Broker architecture applied in Section 3.3 and instantiated for the MT-1 islanding decision in Section 4.3.

1.4 MA Extension Applicability — Four-Filter Assessment

Per DBA-ES-MA-SF v0.2 §1.4 and DBA-ES-MA-QB v0.1 §5, each DBA-ES FC instrument must present a structured four-filter applicability assessment. For FC5, the assessment is resolved at the class level as follows.

Filter 1 — Geolocation relative to defense-critical loads: SATISFIED BY DEFINITION. The FC5 facility class definition (Section 2.2) requires defense-critical or critical service load dependency as the distinguishing characteristic. Every FC5 facility by definition supplies defense installations, emergency operations centers, hospitals, water treatment, or other loads whose interruption produces consequences beyond the facility boundary. Filter 1 is satisfied at class level without further facility-level analysis.

Filter 2 — Position in strategic transmission corridors: SATISFIED. FC5 facilities receive power at HV or EHV (69 kV and above) — the threshold that defines bulk electric system transmission. Each FC5 substation is a node in the sub-transmission or distribution network serving its defense-critical load area. The transmission feed whose failure initiates the MT-1 islanding decision is by definition a strategic supply path for the loads within the islanding boundary.

Filter 3 — Substitutability: LOW. The edge generation asset (BESS, solar+BESS, or diesel/gas backup) is sized to sustain the islanded critical load for 72 hours minimum per the series standard. It has no external substitute during the island interval. The inverter firmware compromise scenario (Scenario A, Section 5.2) demonstrates that the edge generation asset itself can be subverted — meaning the substitutability of the islanding function can be eliminated by the adversary without physical destruction of the equipment.

Filter 4 — Adversary targeting logic: HIGH PRIORITY TARGET. FC5 facilities are the distribution last-mile for defense-critical loads. An adversary targeting the defense-critical load supply chain — whether through the transmission feed (FC2 pattern), the grid control system (CntlCo pattern), or the distribution edge layer (FC5 pattern) — will identify FC5 facilities as the final resilience barrier. Scenario A (inverter firmware compromise at islanding) is the FC5-specific governing MA scenario: it targets the moment of maximum vulnerability, when the facility is transitioning from grid-connected to islanded mode and the Command Broker authorization decision is in progress.

Conclusion: The MA design basis applies to all FC5 facilities. No facility-level exemption assessment is required.

1.5 MA Communications Design Basis

DBA-ES-MA-QB v0.1 §5 establishes the MA communications design basis as a sector-level requirement. The MA communications baseline — primary SCADA and voice communications channels unavailable — is the assumed condition for all MA scenario analysis. The following instantiates CB-1 through CB-5 for FC5, with a note specific to the MT-1 islanding / Command Broker interaction.

FC5-specific note — CB-3 and MT-1 interaction: The most operationally significant CB obligation for FC5 is the intersection of CB-3 (Command Broker continuity) and the MT-1 islanding decision. The MT-1 decision — to island the facility from the transmission feed — is above the Bright Line and requires Command Broker authorization (Section 3.3). Under MA conditions, communications degradation may coincide with the grid stress event that would trigger MT-1. The design basis must ensure that the Command Broker can be reached through the CB-1 backup channel at precisely the moment when the MT-1 authorization is needed. This is the FC5 instantiation of the independent communications channel design basis registered as DI-5 and deferred to the CB-UC instrument.

CB-1 (Backup channel): The FC5 facility must maintain at least one backup communications channel to the Command Broker — the qualified DisCo operator or CntlCo-designated switching authority — that is architecturally independent of the primary SCADA path and the station service bus. The backup channel must remain functional under the Scenario B station service loss condition and under the MA communications baseline condition simultaneously.

CB-2 (Local authority): Substation operators must have pre-authorized authority to execute the FC5 emergency response procedures — including PE-SS response, protective action initiation, and load shedding within the islanded boundary — without CntlCo direction. The MT-1 islanding decision itself requires Command Broker authorization (CB-3) and is not included in the pre-authorized local authority scope. All other emergency actions that do not require above-Bright-Line authorization are within local authority.

CB-3 (Command Broker continuity): MA communications degradation does not suspend the Command Broker obligation for the MT-1 islanding decision or any other above-Bright-Line action at FC5. The qualified DisCo operator or CntlCo-designated authority remains the mandatory Command Broker. The MT-1 default state under communications loss — no islanding without Command Broker authorization — is the fail-safe position. The CB-UC instrument (DI-5) will provide the channel design basis for Command Broker authorization under degraded communications.

CB-4 (Verification cadence): The backup communications channel to the Command Broker must be tested at an interval specified in the facility design basis implementation. The test condition must simulate the MA baseline: primary SCADA and station service communications unavailable, backup channel as sole path to Command Broker.

CB-5 (Degraded-mode procedures): FC5 emergency operating procedures must include a degraded-communications mode. Procedures must address: (a) autonomous facility actions when CntlCo contact is lost (pre-authorized local authority scope); (b) backup channel activation sequence for Command Broker contact; (c) minimum operating state maintained pending communications restoration; (d) threshold conditions under which the facility defaults to grid-connected mode rather than attempting islanding without Command Broker authorization.

Section 2 — Facility Profile: Distribution Substation with Edge Generation

2.1 Facility Class Definition

A facility within the scope of DBA-ES-DI-FC5 must satisfy all three of the following criteria.

Criterion 1 — Voltage transformation: The facility receives power at high voltage (HV) or extra-high voltage (EHV) — 69 kV and above — and transforms it to medium voltage (MV) for distribution — typically 4 to 25 kV — serving a defined service territory.

Criterion 2 — Edge generation: The facility hosts or directly controls on-site generation assets capable of supplying the islanded load. Edge generation includes but is not limited to: utility-scale battery energy storage systems (BESS), solar PV with co-located BESS, fuel cell power conditioning systems, backup diesel or natural gas generation of sufficient capacity to sustain critical islanded load, and hybrid combinations thereof. The edge generation must be sized to sustain the facility’s defense-critical or critical service load for a minimum of 72 hours at rated islanded load.

Criterion 3 — Islanding architecture: The facility’s protection and control architecture supports intentional islanding — the ability to disconnect from the transmission feed and operate in a self-contained mode. The intentional islanding function must be implemented with a Command Broker authorization requirement as specified in Section 3.3. Anti-islanding protection as required by IEEE 1547 must be coordinated with the intentional islanding scheme.

2.2 Defense-Critical Load Dependency

FC5 facilities are distinguished from other distribution substations by their defense-critical or critical service load dependency. The design basis is anchored in the consequence of load supply interruption to defense installations, emergency operations centers, water treatment facilities, hospitals, or other loads whose loss produces consequences beyond the facility boundary. The 72-hour minimum fuel reserve design basis (Section 3.4) is specifically calibrated to defense-critical load dependency: it provides a minimum resilience duration consistent with emergency response mobilization timelines, while acknowledging that defense-critical facilities with longer mission requirements must specify fuel reserves beyond the series minimum.

2.3 Asset Taxonomy

The FC5 asset taxonomy comprises four layers. The physical substation layer includes the HV/EHV transformer(s), switchgear, protection relays, and bus infrastructure. The distribution layer includes the MV feeders, sectionalizing equipment, and load-side protection. The edge generation layer includes all generation assets, inverters, power conditioning units, BESS, and fuel supply infrastructure. The control and communications layer includes the substation SCADA system, microgrid controller, Command Broker communications channel (primary and backup), and the islanding protection scheme.

Section 3 — Series-Significant Architectural Requirements

3.1 Scope of This Section

This section establishes the two series-significant architectural requirements that distinguish FC5 from all other DBA-ES facility class instruments. Both requirements have implications beyond FC5 and are propagated as series standards. Section 3.2 establishes the Bright Line applied to mode transitions. Section 3.3 specifies the Command Broker requirement for intentional islanding. Section 3.4 establishes the minimum fuel reserve design basis.

3.2 The Bright Line Applied to Mode Transitions

The Bright Line is defined in FD-BL v0.2 (§4.2–§4.4) and applied to this facility class by reference to DBA-ES-SF. It is a threshold on individual actions: an action is above the line when its incorrect performance could produce a consequence at or above the facility's severity threshold, and requires Command Broker authorization (FD-BL §4.5). It is not a boundary between technology classes; a deterministic component can originate an above-line action, and determinism governs only the scope carve-out (FD-BL §4.4).

This principle is established here as a series first: mode transitions with cross-boundary consequence potential are above-Bright-Line events requiring Command Broker authorization. This principle is distinguishable from the Bright Line applied to information-processing ML systems (state estimator ML augmentation in DBA-ES-CntlCo, relay settings from ML optimization in DBA-ES-TX-FC2) because the mode transition is not an ML output but an operational command. The Bright Line applies because the decision is non-deterministic — it depends on real-time grid conditions, defense mission status, and operator situational awareness that cannot be fully captured in a deterministic algorithm.

3.3 Command Broker Requirement for Intentional Islanding

Every intentional islanding transition (MT-1) and every intentional grid-reconnection following an islanded period (MT-3) must be authorized by a licensed human Command Broker before execution. The Command Broker for FC5 is the qualified system operator with authority over the facility’s operating mode — typically the DisCo control room operator in coordination with the defense installation energy manager or facility operator.

Authorization channel requirement: The Command Broker must be reachable via a communications channel that is physically independent of the primary SCADA path. The authorization channel must remain functional under the conditions that typically precede an MT-1 decision — transmission fault, grid instability, or planned maintenance switching —

which may degrade or eliminate the primary SCADA communications path. The CB-Framework Cross-Sector v0.1 governs the architecture of the Command Broker communications channel. The FC5-specific instantiation defers the channel design basis to a CB-UC instrument (Open Item DI-5 — see Section 7).

Automated islanding detection: Automated protection schemes that detect islanding conditions and initiate protective tripping (anti-islanding per IEEE 1547) operate below the Bright Line as deterministic protective functions. The Bright Line applies to the intentional mode transition decision, not to protective automatic actions. An unintentional island that forms following a transmission fault and is detected by automatic protection is a PE-1 event (transmission feed loss), not an MT-1 event. MT-1 is initiated only by deliberate Command Broker authorization.

3.4 Minimum Fuel Reserve Design Basis

The minimum fuel reserve design basis for FC5 facilities with on-site fuel-dependent generation (diesel, natural gas, hydrogen fuel cell) is 72 hours at the rated islanded load. This is a series standard, not a site-specific design parameter. Individual FC5 facilities whose defense-critical load mission requires extended islanded operation beyond 72 hours must specify a facility-specific fuel reserve that exceeds this minimum. The 72-hour minimum is the series floor, not the ceiling.

For gas-fired edge generation, the fuel supply design basis under islanded conditions is governed by DBA-GAS-FC4 (Local Distribution: LDC City Gate to End-Use). If the microgrid’s gas-fired generation draws fuel from the same distribution system that may be unavailable during islanded operation, an on-site compressed gas or LNG backup supply must be specified to satisfy the 72-hour minimum. This resolves Open Item DI-3 from v0.1 by reference to DBA-GAS-FC4 §3.2.

3.5 Dual Design Basis Architecture

Consistent with DBA-ES §3.4, an FC5 facility may simultaneously carry a sector-specific physical Design Basis (this document) and an AI Governance Design Basis under the AI Governance series. The Command Broker architecture established in Section 3.3 is itself a Bright Line governance structure applicable to the physical MT events; the AI Governance Design Basis extends this architecture to AI systems within the microgrid control layer.

FC5 facilities with edge generation — solar PV inverters, BESS inverters, and fuel cell power conditioning units — depend on inverter firmware to govern real and reactive power output, islanding detection, and grid-reconnection sequencing. The Bright Line architecture assumes that below-Bright-Line ML systems are deterministic and formally testable. This assumption requires that the firmware governing the inverter’s behavior is known, complete, and free of undisclosed functionality. An inverter whose firmware has not been independently verified cannot be treated as a below-Bright-Line autonomous ML system. This principle was first established in DBA-ES-GN-FC8 §3.3 and is propagated here to the FC5 edge generation context.

AI governance interaction point FC5-AI-1 — Inverter firmware trustworthiness as Bright Line precondition: Inverter firmware verification against an independent source-code-equivalent standard is a precondition for autonomous inverter operation in any islanded or grid-reconnection application. An inverter operating on unverified firmware must be treated as operating above the Bright Line, requiring Command Broker oversight for islanding mode transitions and grid-reconnection sequencing, consistent with the MT-1 Command Broker requirement in Section 3.3.

AI governance interaction point FC5-AI-2 — Microgrid controller ML optimization: Microgrid controller ML systems that optimize generation dispatch, load shedding priorities, and storage state-of-charge within the island may operate below the Bright Line when performing deterministic optimization within trained envelopes. They may not initiate mode transitions autonomously. Any ML-generated recommendation to initiate MT-1 or MT-3 must be presented to the Command Broker as advisory only.

AI governance interaction point FC5-AI-3 — Gen AI operator decision support: Gen AI tools that assist the DisCo control room operator in evaluating islanding decisions — assessing load priority, generation availability, and grid conditions — operate above the Bright Line. The licensed system operator is the mandatory Command Broker. Gen AI output on islanding decisions is advisory only; the authorization must come from the human Command Broker.

MA-1/AI governance attack surface convergence (FC5 instantiation): The inverter firmware compromise scenario (Section 5, Scenario A) establishes that the adversarial ML attack surface and the physical consequence event converge at the same point: the inverter firmware. A supply chain compromise that introduces undisclosed firmware functionality is simultaneously an adversarial ML attack (the firmware behaves as an ML system whose training data has been tampered with) and a physical consequence event (the inverter trips or destabilizes the island at the moment of MT-1 execution). Inverter firmware verification satisfies both the AI Governance Design Basis and the physical Design Basis. This is the second instantiation of the MA-1/AI governance convergence principle, first established in DBA-ES-CntlCo v0.2 §7.1.3 and propagated to DBA-ES-TX-FC3 v0.2 §3.6.

Section 4 — Design Basis: FC5 Postulated Event Categories

4.1 Sector-Level Inheritance and Locally Defined Category

FC5 inherits four sector-level postulated event categories from DBA-ES §3.5: PE (Power Supply and Grid Interface), SE (Structural and Physical Integrity), CE (Control and Communications), and XC (Cross-Sector Cascade). A fifth category — MT (Mode Transition Events) — is locally defined for FC5, following the precedent of the DBA-DC series' WM (Workload Migration) category. MT events are unique to the islanding-capable architecture of the FC5 facility class and have no analog in any other DBA-ES facility class instrument.

4.2 Category PE — Power Supply and Grid Interface Events

PE events address loss or degradation of the HV/EHV transmission feed and the generation supply available to the island. Three sub-cases govern the PE design basis for FC5.

PE-1 — Transmission feed loss: Complete loss of the HV/EHV transmission feed, requiring transition to islanded operation or loss of all supply. The PE-1 bounding case is a sustained transmission feed loss — exceeding the duration of available spinning reserve and islanded generation capacity — during a period of peak defense-critical load demand.

PE-2 — Fuel supply exhaustion: Depletion of on-site fuel supply for fuel-dependent edge generation during sustained islanded operation. The 72-hour fuel reserve design basis bounds the PE-2 consequence duration. For gas-fired edge generation, the fuel supply design basis under islanded conditions is resolved by reference to DBA-GAS-FC4 §3.2 (Open Item DI-3 closed).

PE-3 — Generation capacity shortfall: Edge generation capacity insufficient to serve the full islanded critical load at rated power, requiring load shedding within the island. The PE-3 bounding case is a single largest edge generation unit trip during islanded operation, requiring immediate load shedding to prevent frequency collapse within the island.

PE-SS — Station Service Loss (series-significant sub-case): Loss of station service power supply to protection, control, and communications equipment at the substation. Three sub-cases: PE-SS-1 (station service battery exhaustion following extended AC supply loss), PE-SS-2 (station service transformer failure), PE-SS-3 (dual-feed station service loss during islanded operation when both station service feeds are derived from the islanded bus). PE-SS-3 is the worst-case sub-case: loss of station service during islanded operation disables protection, SCADA, and Command Broker communications simultaneously.

4.3 Category MT — Mode Transition Events (Locally Defined)

MT events address the transitions between grid-connected and islanded operating modes, and the failure mode that leaves the facility in an indeterminate state. The Command Broker requirement for MT-1 and MT-3 is a binding architectural requirement established in Section 3.3; it applies to every MT-1 and MT-3 event without exception.

MT-1 — Intentional islanding (Command Broker authorized): The facility transitions to islanded operation under Command Broker authorization. The bounding design basis for MT-1 is execution under degraded conditions: transmission fault active, primary SCADA path unavailable, edge generation mix below optimal. The Command Broker must evaluate the transition with incomplete information and maintain authorization authority over the backup communications channel.

MT-2 — Unintentional island formation: The facility forms an unintentional island following a transmission fault, tripped by automatic protection before Command Broker authorization can be obtained. MT-2 is a protective response to a PE-1 event. The island may be unsustainable if edge generation is insufficient or fuel reserves are not at design levels. The MT-2 bounding case is an unintentional island that forms at maximum load with minimum generation.

MT-3 — Intentional grid-reconnection (Command Broker authorized): The facility reconnects to the bulk grid following an islanded period. MT-3 requires synchronization check, relay confirmation, and Command Broker authorization. The MT-3 bounding case is a reconnection attempt following a prolonged islanded period during which grid frequency and angle at the point of common coupling are unknown to the microgrid controller.

MT-4 — Failed mode transition (indeterminate state): A mode transition that is initiated but not completed, leaving the facility in an indeterminate state — neither fully islanded nor fully grid-connected. MT-4 is the most dangerous MT event category because protection relay coordination assumptions may be violated in the indeterminate state. The MT-4 bounding case is a breaker failure during MT-1 execution that leaves one phase connected and two phases isolated.

4.4 Category SE — Structural and Physical Integrity Events

SE events address physical damage to facility infrastructure that degrades or eliminates the facility's ability to serve the islanded load.

SE-1 — Distribution transformer failure: Loss of the main step-down transformer serving the distribution bus. SE-1 is the single highest-RT_HVLLC event for FC5: distribution transformer replacement timelines are 12–24 months for custom units. The SE-1 bounding case is a transformer failure with no spare transformer available and a replacement lead time that exceeds the duration of any credible emergency workaround.

SE-2 — Switchgear failure: Loss of primary bus switchgear, preventing transfer to alternate supply or restoration of supply to the distribution bus. The SE-2 bounding case is a switchgear fault that requires deenergization of the main bus, preventing islanded operation until repair is complete.

SE-3 — Physical security breach: Unauthorized physical access to substation equipment or edge generation infrastructure. The SE-3 design basis event is a physical attack that disables the Command Broker communications channel and the primary SCADA path simultaneously, preventing both manual and automated mode transition control.

4.5 Category CE — Control and Communications Events

CE events at FC5 operate across three distinct control layers: the substation SCADA and protection system, the microgrid controller, and the Command Broker communications channel. Events affecting any one layer have mode-dependent consequences. CE-1, during islanded operation, blocks the Command Broker's ability to authorize MT-3, creating a potential isolation trap: the facility cannot resynchronize without Command Broker authorization, and the Command Broker cannot be reached because CE-1 has removed the SCADA communications path. The independent authorization channel required by the MT-1/MT-3 Command Broker requirement (Section 3.3) directly addresses this scenario.

CE-1 — SCADA communications loss: Loss of primary SCADA communications path. Mode-dependent: during grid-connected operation, CE-1 forces manual operation of MT-1 through the backup channel; during islanded operation, CE-1 blocks MT-3 unless the independent authorization channel remains functional.

CE-2 — Microgrid controller failure: Loss of the microgrid controller that manages generation dispatch, load shedding, and frequency regulation within the island. CE-2 during islanded operation may result in frequency instability if generation dispatch is not manually assumed.

CE-3 — Protection relay malfunction: Incorrect operation of protection relays, including nuisance tripping, failure to trip, or coordination failure under islanded conditions. Protection relay settings optimized for grid-connected operation may not provide adequate protection under islanded conditions where fault current levels differ from grid-fed levels.

4.6 Category XC — Cross-Sector Cascade Events

XC-1 — Defense-critical load interruption: Loss of supply to a defense installation within the FC5 islanding boundary. The XC-1 consequence triggers the Cross-Sector Consequence Elevation Rule per DBA-EN-SF §3.4 when the defense installation is the sole supply path for a Tier 2 or Tier 3 DBA-designated function. The XC-1 design basis requires identification of defense-critical loads and their mission-duration requirements.

XC-2 — Critical service cascade: Loss of supply to hospitals, water treatment facilities, or emergency communications infrastructure within the service territory. XC-2 is a Tier 2 consequence when the affected critical services serve a geographic area whose loss produces a grid-level or cross-sector consequence.

XC-3 — Reverse cascade from failed resynchronization: A failed MT-3 (grid-reconnection) that injects out-of-phase power into the transmission system, causing protection relay trips at adjacent FC2 or FC3 facilities. XC-3 is the primary cross-boundary consequence of a failed mode transition and is the basis for placing MT-3 above the Bright Line. The XC-3 bounding case is a reconnection attempt during a system-wide frequency disturbance when the island frequency has diverged from nominal.

Section 5 — Design Basis Accident Scenarios

5.1 Governing Bounding Event

The DBA-MA bounding event for the FC5 facility class is a coordinated inverter firmware compromise executed at the moment of islanding transition. This attack converts the edge generation fleet from a resilience asset into a grid-destabilizing instrument at the precise moment the facility depends on it most. The attack exploits the supply chain vulnerability established in DBA-ES-GN-FC8 §3.2, instantiated at the distribution-edge scale where the inverter fleet is smaller, less monitored, and more diverse in vendor origin than utility-scale IBR clusters.

The four MA characteristics apply as follows: Intentionality — the firmware is pre-positioned through supply chain compromise; the triggering event is the islanding command itself, converted from a resilience action into a weapon trigger. Persistence — the firmware compromise persists undetected across normal inverter operation until triggered; standard operational testing does not reveal the compromised behavior because it is activated only by the specific sequence of islanding detection followed by Command Broker authorization. Precision — the attack targets the specific inverter models deployed in the defense-critical FC5 facility; the adversary has conducted reconnaissance on the facility's edge generation asset mix. Denial — the island forms black; defense-critical loads lose power at the moment they are most dependent on the islanded supply.

5.2 Scenario A — Inverter Firmware Compromise at Islanding (Governing MA Scenario)

5.2.1 Initiating Event

A coordinated supply chain attack pre-positions compromised firmware in BESS inverters and solar PV inverters at an FC5 facility serving a defense installation. The firmware operates normally under grid-connected conditions. The compromise is activated by the specific sequence: islanding detection signal received, followed within 30 seconds by Command Broker authorization for MT-1 execution. Upon activation, the compromised firmware commands simultaneous real power withdrawal and reactive power injection from all affected inverters, producing a voltage spike and frequency collapse within the forming island.

5.2.2 Consequence Chain

The island attempts to form following a transmission fault. The Command Broker authorizes MT-1. The main breaker opens, disconnecting the facility from the bulk grid. Within 30 seconds of MT-1 execution, the compromised inverters respond to the islanding detection signal by simultaneously withdrawing real power output. Island frequency collapses below the inverter under-frequency trip setting. All inverters trip on frequency protection. The island forms black: zero generation available, defense-critical loads de-energized.

The consequence is Tier 1 at the facility level (total loss of supply to the islanded load) and Tier 2, where the defense installation load meets the Cross-Sector Consequence Elevation Rule criteria of DBA-EN-SF §3.4. Recovery requires physical inspection and firmware verification of all inverters before the island can be re-formed — a process that requires vendor engagement, independent firmware verification, and replacement of non-conforming units. Recovery timeline: days to weeks, depending on spare inverter availability and firmware verification capability.

5.2.3 MA-1/AI Governance Convergence

The Scenario A attack surface is simultaneously an adversarial ML attack (the inverter firmware is an ML-equivalent control system whose supply chain integrity has been compromised) and a physical consequence event (the island forms black). Inverter firmware verification satisfies both the AI Governance Design Basis (FC5-AI-1: firmware trustworthiness as Bright Line precondition) and the physical Design Basis (islanding architecture integrity). This is the FC5 instantiation of the MA-1/AI governance convergence principle established in DBA-ES-CntlCo v0.2 §7.1.3.

5.3 Scenario B — Station Service Loss Cascade Under Islanded Conditions (Non-Adversarial DBA Bounding Scenario)

5.3.1 Initiating Event

During sustained islanded operation (the facility has been islanded for 18 hours following a transmission fault), the station service transformer fails. Both station service feeds derive from the islanded bus. Loss of station service simultaneously removes AC power from protection relay panels, SCADA communications equipment, the microgrid controller UPS, and the Command Broker communications channel. Battery backup for protection panels exhausts within 4 hours.

5.3.2 Consequence Chain

Microgrid controller loses AC supply; UPS sustains operation for up to 2 hours. SCADA communications path drops immediately. Command Broker communications channel — if dependent on SCADA infrastructure for backhaul — drops simultaneously. The DisCo operator loses all visibility into the islanded facility. The Command Broker cannot authorize MT-3 because the authorization channel is unavailable. The island continues operating under the last-known microgrid controller state until UPS exhaustion or a generation event forces a response. If the Command Broker cannot be reached via the independent authorization channel (per the CB-Framework requirement), the islanded facility is effectively autonomous, operating without oversight for the duration of station service restoration.

This is the non-adversarial bounding event that demonstrates why the independent Command Broker communications channel (Section 3.3) must be physically independent of the station service bus as well as the primary SCADA path. A PE-SS-3 event removes both the primary and secondary control paths if both derive from the islanded bus.

5.4 Scenario C — Coordinated Feed Attack and Gas Supply Interdiction (Compound MA Scenario)

5.4.1 Initiating Event

A coordinated attack simultaneously destroys the HV/EHV transmission feed transformer serving the FC5 facility (SE-1 physical attack). It disrupts gas supply to the local distribution system feeding the facility’s gas-fired edge generation (per DBA-GAS-FC4 governing event). The attack is executed during a period of maximum defense-critical load demand — a winter emergency with heating load at peak.

5.4.2 Consequence Chain

The transmission feed transformer loss triggers MT-2 (unintentional island formation). The island forms with diesel BESS and solar PV available; gas-fired edge generation is unavailable because the gas distribution pressure has fallen below the minimum inlet pressure for the gas turbines. The island operates on reduced generation capacity. Fuel oil BESS cycling and solar PV (weather permitting) sustain the island at reduced load through load shedding. The 72-hour fuel reserve design basis for diesel edge generation bounds the survival horizon.

The compound consequence: transformer replacement timeline (12–24 months) exceeds any credible fuel reserve duration. The facility cannot restore full supply without either a spare transformer or a temporary generation interconnection — neither of which is immediately available in the DBA-MA scenario, where the adversary has targeted the transformer specifically for its RT_HVLLC consequence. Gas supply restoration (per DBA-GAS-FC4 recovery architecture) provides partial relief when gas distribution pressure is restored, but the transformer loss remains the governing constraint. The consequence is Tier 2 — extended defense-critical load curtailment lasting the transformer replacement timeline.

Section 6 — Recovery Architecture

6.1 Island Restoration Sequencing

Recovery from a black island condition (Scenario A) follows a defined sequencing requirement before generation can be restarted and the island re-formed. The sequence is: (1) physical inspection of all inverters for signs of tampering or unauthorized maintenance access; (2) firmware version verification against the facility's inverter firmware registry; (3) independent firmware verification against the source-code-equivalent standard for any inverter whose on-site version does not match the registry; (4) replacement of any inverter that fails verification; (5) black-start of verified generation units in generation-priority order; (6) load restoration in critical-load priority order; (7) Command Broker authorization of MT-3 for grid-reconnection when grid conditions permit.

Step 3 (independent firmware verification) is the governing constraint on the recovery timeline. Vendor-supplied firmware verification tools that do not provide access to source-code-equivalent comparison are not sufficient for DBA-MA recovery purposes. The facility must have a pre-established firmware verification protocol and relationship with the inverter vendor before an event occurs.

6.2 Grid-Reconnection Authorization

MT-3 (grid-reconnection) requires synchronization check relay confirmation that the island and bulk grid are within the synchronization window (typically ±0.1 Hz frequency difference and ±20° angle difference) before the reconnecting breaker is permitted to close. The Command Broker must confirm: (1) synchronization check relay indicates within-window condition; (2) independent firmware verification is complete for all inverters that will operate in grid-connected mode; (3) the bulk grid operator (CntlCo) has been notified of the planned reconnection. The CB-Framework Cross-Sector v0.1 governs the Command Broker authorization protocol.

6.3 Transformer Replacement Recovery (Scenario C)

SE-1 (transformer failure or destruction) produces the longest FC5 recovery timeline. The governing recovery architecture requires: (1) immediate activation of the strategic transformer reserve notification protocol (per RT_HVLLC Strategic Reserve White Paper v0.2); (2) engagement with the applicable RTO/ISO for emergency generation interconnection to sustain defense-critical load during the replacement period; (3) fuel supply management for diesel edge generation to extend the survival horizon beyond the 72-hour minimum; (4) coordination with DBA-GAS-FC4 recovery architecture for gas supply restoration to gas-fired edge generation.

Section 7 — Open Items and Deferred Items

Item	Description	Resolution Criterion
OI-1 CLOSED	DBA-GAS §8.3 and DBA-OIL §8.3 Bright Line naming correction. Both documents used “Bright Line” to mean federal coordination triggers. Resolved: DBA-GAS-UC-LNG v1.1 §6.3 heading renamed to “Federal Coordination Thresholds.” DBA-GAS-FC1 correction deferred as a separate OI.	Closed. Partially resolved; GAS-FC1 correction carried as a separate item.
DI-1 CLOSED	DBA Consequence Envelope (v0.2). Full DBA non-adversarial consequence envelope for FC5. Resolved in this instrument: Section 5 scenarios B and C establish the non-adversarial bounding events and consequence chains.	Closed. DBA consequence envelope complete.
DI-2 CLOSED	MA Extension (v0.2). Military action extension for FC5. Resolved in this instrument: Section 5 Scenario A establishes the governing MA bounding event (inverter firmware compromise at islanding). Section 3.5 establishes MA-1/AI governance convergence.	Closed. MA extension complete.
DI-3 CLOSED	Gas supply design basis for gas-fired edge generation under islanded conditions. Resolved by reference to DBA-GAS-FC4 §3.2 (Local Distribution: LDC City Gate to End-Use). If microgrid gas-fired generation draws from the distribution system that may be unavailable during islanding, on-site compressed gas or LNG backup supply is required per DBA-GAS-FC4.	Closed. Resolved by reference to DBA-GAS-FC4 v0.1 §3.2.
DI-4	Service area defense-critical and critical load inventory. XC-1 and XC-2 require identification of defense-critical loads and critical services within the FC5 islanding boundary. This inventory is facility-specific and must be performed by the facility operator for each FC5 installation.	Critical load inventory completed by each FC5 facility operator as part of site-specific design basis implementation.
DI-5	Command Broker independent communications channel design basis. The FC5-specific design basis for the backup authorization channel — physically independent of both the primary SCADA path and the station service bus (per Scenario B) — is deferred to a CB-UC instrument.	Deferred to the CB-UC instrument. CB-Framework Cross-Sector v0.1 governs the architecture.
DI-6 CLOSED	Dual-DB AI Governance interaction points. Bright Line boundary for FC5 microgrid control AI applications. Resolved in this instrument: Section 3.5 establishes three FC5-specific	Closed. AI governance interaction points established in §3.5.

	AI governance interaction points (FC5-AI-1, FC5-AI-2, FC5-AI-3) and the MA-1/AI governance convergence for the inverter firmware scenario.	
OI-2	Inverter firmware verification protocol. Section 3.5 (FC5-AI-1) and Section 6.1 establish inverter firmware verification as a precondition for autonomous islanded operation and as a recovery requirement following Scenario A. The specific verification protocol — the technical standard against which firmware is verified and the vendor engagement model — is not established at the series level. This requires development in a companion implementation guide or DBA-VC vendor companion update.	Inverter firmware verification protocol established in DBA-VC or companion implementation guide.
DI-7 CLOSED	MA applicability four-filter block and MA communications design basis (N-15/N-16 batch pass).	CLOSED v0.4. Four-filter MA applicability structured assessment added as §1.4 per DBA-ES-MA-SF v0.2 §1.4 and QB v0.1 §5; all four filters satisfied at class level. MA communications design basis CB-1 through CB-5 added as §1.5 with FC5-specific CB-3/MT-1 islanding interaction note. Closes N-15/OI-8 and N-16/OI-1 for FC5.

Section 8 — References

Regulatory and Standards References

Institute of Electrical and Electronics Engineers. IEEE 1547-2018: Standard for Interconnection and Interoperability of Distributed Energy Resources with Associated Electric Power Systems Interfaces. New York: IEEE, 2018.

North American Electric Reliability Corporation. PRC-024-3: Generator Frequency and Voltage Protective Relay Settings. Atlanta: NERC. Current version.

North American Electric Reliability Corporation. CIP-014-3: Physical Security. Atlanta: NERC. Current version.

Federal Energy Regulatory Commission. Order No. 2222: Participation of Distributed Energy Resource Aggregations in Markets Operated by Regional Transmission Organizations and Independent System Operators. Washington, D.C.: FERC, 2020.

Government and Policy References

Defense Science Board Task Force on DoD Energy Strategy. “More Fight — Less Fuel.” Washington, D.C.: Office of the Under Secretary of Defense for Acquisition, Technology and Logistics, February 2008.

Department of Energy. “QRI Assessment: Quadrennial Energy Review.” Washington, D.C.: DOE, 2015. (Grid resilience and distributed energy resources.)

Technical References

Lasseter, R.H. “MicroGrids.” Proceedings of the 2002 IEEE Power Engineering Society Winter Meeting. New York: IEEE, 2002. (Foundational reference for microgrid islanding architecture.)

Kroposki, B. et al. “Achieving a 100% Renewable Grid: Operating Electric Power Systems with Extremely High Levels of Variable Renewable Energy.” IEEE Power and Energy Magazine, March/April 2017. (Frequency response of inverter-based islanded systems.)

Series Parent and Cross-Reference Documents

Roxey, Tim. DBA-EN-SF: Energy Sector Design Basis Framework. Version 0.8 — DRAFT. Eclectic Technologies, June 2026.

Roxey, Tim. DBA-ES: Electric Sector Design Basis Framework. Version 0.9 — DRAFT. Eclectic Technologies, June 2026.

Roxey, Tim. DBA-ES FC5 and CntlCo Gap Analysis. Version 0.2 — DRAFT. Eclectic Technologies, June 2026. (Predecessor document.)

Roxey, Tim. DBA-ES-TX-FC2: Large EHV Transmission Substation. Version 0.2 — DRAFT. Eclectic Technologies, June 2026.

Roxey, Tim. DBA-ES-GC-FC4: Natural Gas Compressor Station. Version 0.4 — DRAFT. Eclectic Technologies, June 2026.

Roxey, Tim. DBA-ES-GN-FC8: Renewable Generation and Battery Energy Storage. Version 0.1 — DRAFT. Eclectic Technologies, June 2026. (Inverter firmware trustworthiness Bright Line precondition established in §3.3.)

Roxey, Tim. DBA-GAS-FC4: Local Distribution (LDC City Gate to End-Use). Version 0.1 — DRAFT. Eclectic Technologies, June 2026. (Gas supply design basis for islanded gas-fired edge generation — DI-3 resolution.)

Roxey, Tim. DBA-ES-CntlCo: Electric Sector Control-Company Functional Class. Version 0.3 — DRAFT. Eclectic Technologies, June 2026. (MA-1/AI governance convergence principle established in §7.1.3; SABD T_1/T_2 calibration in §3.4.)

Roxey, Tim. DBA-ES-MA-SF: Electric Sector Military Action Sector Framework. Version 0.2 — DRAFT. Eclectic Technologies, June 2026.

Roxey, Tim. DBA-ES-MA-QB: Electric Sector Military Action Quantitative Design Basis. Version 0.1 — DRAFT. Eclectic Technologies, June 2026.

Roxey, Tim. CB-Framework: Command Broker Cross-Sector Framework. Version 0.1 — DRAFT. Eclectic Technologies, June 2026.

Roxey, Tim. Command Broker Implementation Guide. Version 0.1 — DRAFT. Eclectic Technologies, June 2026. (Bright Line and Command Broker architecture reference.)

Roxey, Tim. RT_HVLLC Strategic Reserve White Paper. Version 0.2 — DRAFT. Eclectic Technologies, June 2026. (Transformer replacement timeline and strategic reserve notification protocol.)

Revision History

Version	Date	Author	Description
0.1	June 2026	T. Roxey	Initial working draft. DB-layer instrument only; DBA consequence envelope and MA extension deferred to v0.2. Inherits postulated event architecture from DBA-ES FC5 and CntlCo Gap Analysis (v0.1) as predecessor document. Established facility class definition (HV/EHV input; 69–110 kV output; edge generation; islanding capability). Developed a full postulated event taxonomy, including all inherited sector-level categories (PE, SE, CE, XC) and the locally defined MT (Mode Transition) category. Incorporated PE-SS (Station Service Loss) with three sub-cases as a series-significant postulated event. Established the Command Broker requirement for intentional islanding (MT-1) as a Bright Line architectural requirement. Established a 72-hour minimum fuel reserve at the rated islanded load as a series standard for FC5 facilities. Acknowledged dual-DB architecture per DBA-ES §3.4. Registered DBA-GAS §8.3 and DBA-OIL §8.3 Bright Line naming correction as Open Item OI-1.
0.2	June 2026	T. Roxey	Full three-layer instrument. DBA consequence envelope added: Section 3.2 establishes the DBA-MA bounding event (coordinated inverter firmware compromise triggered at islanding). MA extension added: Section 4 scenarios A, B, and C. Section 6 (Recovery Architecture) added. Open items DI-1, DI-2, DI-3, and DI-6 have been closed. DI-5 deferred to the CB-UC instrument. New open item OI-2 registered (inverter firmware verification protocol). Cover updated to v0.3. Section 7 (Open Items) updated. Section 8 (References) updated: DBA-ES-SF v0.9, DBA-GAS-FC4 v0.1, CB-Framework v0.1, DBA-ES-GN-FC8 v0.1 added.
0.4	June 2026	T. Roxey	N-15/N-16 batch pass (Tier B). Three changes: (1) DBA-EN-SF grandparent updated from v0.7 to v0.8 throughout. (2) §1.4 MA Applicability four-filter structured assessment added — all four filters satisfied at class level; FC5 identified as distribution last-mile for defense-critical loads; Scenario A (inverter firmware at islanding) noted as FC5-specific Filter 4 rationale. Closes N-15/OI-8 for FC5. (3) §1.5 MA Communications Design Basis added — CB-1 through CB-5 instantiated with FC5-specific CB-3/MT-1 islanding interaction note: CB-2 backup channel required at precisely the moment of the MT-1 Command Broker authorization decision. Closes N-16/OI-1 for FC5. DI-7 added and marked CLOSED. CntlCo reference updated to v0.3. DBA-ES-MA-SF v0.2 and DBA-ES-MA-QB v0.1 added to references.